



Figure 1: Characteristics of cognitive systems

Ways to modernise your web application

Some of the common patterns to modernise your application with cognitive services are:

- Instead of using textual data as input, use scanning or optical character recognition techniques.
- Instead of written input, leverage speech technologies. The speech-to-text converter will convert into the required text and process.
- For authentication, instead of passwords, use fingerprints or retina scanning.
- For user identification, instead of using standard ID (government ID, driving licence, etc), use facial recognition.
- For registration, instead of using textual input, a document (PDF, Word, etc) can be uploaded and the application can intelligently read the content and auto-fill the details.
- Social media companies use a number of ML/AI algorithms. Integrate them into your web application.

Key benefits of cognitive services

There are a number of benefits of using cognitive services including:

- Intelligent user identification for security, surveillance systems
- Intelligent content moderation by showing only appropriate content in web pages/social media sites
- Cater to diverse audiences by embedding multi-lingual support in the application
- Speaker recognition in web applications for authentication
- Auto suggestions and intelligent understanding of the user and environment to provide relevant and

curated information

- Personalisation in conversational apps by intelligently navigating the user interactions and guiding towards the intent (goal) of the interaction

Challenges of cognitive services

With every great technology comes a bunch of challenges as well. Some of the pitfalls of excessive and unregulated cognitive services include:

- Prevalence of fake news in media applications
- Wilful polarisation of readers in social media sites
- Security risks
- Increase in carbon footprint because of higher computational needs
- Longer development and deployment cycles

Python libraries for cognitive services

You can take advantage of Python libraries to seamlessly integrate cognitive services into your web application. These services are typically exposed using Restful APIs and programming SDKs. Examples include:

- OpenCV for computer vision
- SpeechRecognition and PyAudio for speech to text
- TTS library for text to speech

With cognitive services, the capabilities of web applications are endless. These intelligent algorithms can use natural methods of interaction and make the customer experience better. However, with great power comes great responsibility. It is the responsibility of the provider and creator to use these cognitive capabilities in an ethical way. **END** 🐧

References

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